

BRICS EGEC WG - 2026 REPORT – CONTENTS: CIRCULAR CARBON ECONOMY & VCM

1 A) Circular Carbon Economy - concept note - **INDIA**

1 B) VCM for BRICS - concept note – **INDIA**

2 A) BRICS Voluntary Carbon Hub – **RUSSIA**

2 B) BRICS Energy Attributes Certificates – **RUSSIA**

2C) VCM comments & case study of JSE Venture VCM platform – **SOUTH AFRICA**

3) India Carbon Market - CCTS & VCM - **INDIA**

CIRCULAR CARBON ECONOMY

Since the advent of fossil fuels, Carbon followed a linear path: extraction, conversion to energy and release of CO₂ and other GHG emissions into the atmosphere. This model powered industrialisation, global trade and rising standards of living. It also led to increases in atmospheric stock of CO₂ and other GHG emissions, with consequential adverse impact of Climate Change

Circular Carbon Economy is a framework for carbon cycle, under circular economy framework, viz Reduce Reuse, Recycle and Remove. Rather than relying solely on eliminating emissions the traditional "linear" model

DEFINITION OF CIRCULAR CARBON ECONOMY (CCE) : as per the UNFCCC Circular Carbon Economy page” is summarized below.

Climate change mitigation focused mainly on faster deployment of renewables and replacement of hydrocarbons in the power and transportation sectors. However, global emissions are still growing and greenhouse gas (GHG) concentrations are still at high levels. It seems the current approach is not working.

Mitigation strategies should consider practical and effective solutions that create a holistic approach and make use of all the opportunities for solving this issue. This approach would

accommodate a diverse energy mix of renewable and traditional fuel sources, would lessen atmospheric levels of GHGs, while also sustaining economic growth and prosperity.

As important as renewables are, and as much progress as they have made in recent years, few analysts believe that the world can achieve the Paris Agreement's goal of a balance between sources and sinks by 2050 through renewables alone. The global path toward a carbon balance will inevitably include fossil fuels, and their carbon emissions must be managed.

The concept of the circular carbon economy, an outgrowth of the idea of the circular economy, is a useful framework for understanding how all carbon mitigation options can be linked together in a system that achieves the climate goals laid out in the Paris Agreement.

CCE OFFERS A NEW WAY OF APPROACHING CLIMATE GOALS that implicitly values all options and encourages all efforts to mitigate carbon accumulation in the atmosphere. The circular carbon economy extends the concept of a circular economy (reduce, reuse, recycle) by including remove and focusing exclusively on carbon and energy flows.

CCE is a framework that manages carbon the way a traditional economy manages resources. Instead of treating carbon as just a waste product or a one-time resource, the CCE follows four core strategies (known as the "4 Rs") to close the loop on emissions:

Reduce: Cut down the amount of carbon emitted in the first place through energy efficiency and renewables.

Reuse: Capture emissions and put them directly to use without breaking them down, such as in enhanced oil recovery or certain manufacturing processes.

Recycle: Transform waste carbon dioxide CO₂ into valuable products like synthetic fuels, concrete, plastics, or fertilizers.

Remove: Pull carbon directly out of the air using nature-based solutions (like planting trees) or engineered technologies, then store it underground.

SPECIFIC TECHNOLOGIES IN THE CIRCULAR CARBON ECONOMY (CCE) that drive the transformation of carbon from an environmental liability into an industrial asset. These technologies align with the framework's core pillars:

1. Reduce (Efficiency & Substitution):

Direct Reduced Iron (DRI): Uses green hydrogen or natural gas instead of coal to extract iron from ore, cutting steel industry emissions.

Allam Cycle Power Plants: An advanced oxy-fuel combustion process that burns natural gas with pure oxygen, capturing 100% of the produced carbon dioxide (CO₂) inherently as part of the generation process. The Allam Cycle uses high-pressure, supercritical carbon dioxide (sCO₂) as the fluid that spins the turbines to generate electricity. After spinning the turbine, most of the CO₂ is compressed, reheated, and sent right back to the combustor to keep the cycle going.

2. Reuse (Direct Utilization):

Enhanced Oil Recovery (EOR): Injects captured, compressed CO₂ gas into maturing oil fields. The gas swells the trapped oil, making it easier to extract while trapping the CO₂ underground.

Supercritical CO₂ Working Fluids: Uses CO₂ in a high-pressure, semi-liquid state to replace standard refrigerants in industrial cooling systems.

3. Recycle (Conversion to Products)

In-Situ Mineralization: CO₂ in flue gases (Hydrogen generators, Gas Engines, Cement Kilns, Steel Plants) is mineralized to useful chemical compounds, which leads to permanent storage of CO₂ (eg CaCO₃)

Ex-Situ Mineralization: Reacts captured CO₂ with industrial waste (eg steel slag or fly ash) to produce synthetic concrete aggregates and solid building blocks, permanently trapping the gas.

Power-to-X (P2X) / Synthetic Fuels: Combines captured CO₂ with green hydrogen via a catalyst to synthesize e-methanol, methane, and Sustainable Aviation Fuel (SAF).

Electrochemical CO₂ Reduction: Uses electricity and single-atom catalysts to directly split CO₂ molecules, assembling them into chemical building blocks like polycarbonates and olefins used for making plastics.

4. Remove (Direct Extraction & Storage):

Direct Air Capture (DAC): Employs giant, fan-powered systems with liquid amine solvents or solid chemical sorbents to pull diffuse CO₂ straight out of ambient air.

Biomass-based approaches. Pyrolysis of converts biomass carbon into a stable form, Biochar, which can sequestered in ground (agriculture fields, pits caused by soli extraction, etc) for permanent storage of carbon dioxide.

Bioenergy with Carbon Capture and Storage (BECCS): Burns fast-growing biomass (like wood pellets or agricultural waste) for electricity, then captures and buries the resulting emissions underground.

Enhanced Rock Weathering (ERW): Spreads finely crushed silicate rocks (like basalt) onto agricultural soils. The rock chemically reacts with rainwater to pull CO₂ from the atmosphere, turning it into stable bicarbonate minerals

The value of removal depends on permanence. Temporary storage, long-lived products and geological storage are not equivalent. Transparent standards, independent verification and monitoring systems are critical to credibility.

CARBON CAPTURE UTILIZATION & STORAGE (CCUS): represents a transformative approach with the capacity to create closed loop systems where CO₂ emissions can be captured, stored or converted into value-added products, thereby reducing the dependency on fossil fuels and minimizing waste.

CCUS technologies within the context of circular economy, encompass diverse carbon sources, viz fossil-fuel derived carbon, biomass and atmospheric. Multiple carbon capture technologies are available, with varying technology readiness level and economic viability. Likewise, various carbon storage options in geological formations, sequestration in agriculture fields and forests, along with direct and indirect utilization routes.

Policies, such as carbon tax, carbon markets and fiscal enablers favouring the implementation of CCUS into the circular economy are essential for commercialization of the technologies, proven at Lab or Pilot Plants stages. There is need for significant technology advancements to deploy these technologies at scale, as required to meet net-zero emissions targets,

Moreover, consolidated frameworks for assessment and monitoring, policy support, and international cooperation hold the potential to advance the adoption of this technology within a circular economy.

SHARED FRAMEWORK: The Circular Carbon Economy provides a shared framework through which these pathways can be aligned, measured and scaled. It supports national priorities such as energy security, industrial growth and rural development, while enabling international cooperation.

Opportunities include shared methodologies, common standards, joint research, demonstration projects and investment platforms and the development of markets for verified low-carbon products

BRICS – VOLUNTARY CARBON MARKET

1) The EGEC WG focus, related to Climate, would relate to Clean Energy Fund, under aegis of NDB, initiated by EGEC at level of BBC with NDB, during 2021, which needs to be pursued

2) The other aspect would be Voluntary Carbon Markets (VCM), for credits from projects outside scope of NDC's. Russia prepared framework in 2024 and Brazil carried it forward in 2025.

The expected outcomes being voluntary carbon offset trading initiative among BRICS countries include more cost-effective emissions reductions by enabling countries and companies to purchase offsets from regions with lower mitigation costs.

To bring it to logical conclusion in 2026, we need to have the right platform for VCM and offset trading. It will be Industry and Businesses, who would be undertaking Carbon Offsets trading.

3) The **Financial Services Working Group (FSWG)** of BRICS Business Council has **ESG and Green Finance Subcommittee**.

EGEC WG will make request to ESG and Green Finance Subcommittee to recommend “financial instruments” that support the implementation of CDR and CCUS projects, as well as similar enablers for intra BRICS Carbon Trading under “Voluntary Carbon Markets”.

As **EGEC WG**, we will propose a “**VCM Network**” and onboard Members from BRIC nations, who are interested in developing “carbon offsets” projects as well as Industries which are interested in buying such credits.

4) Carbon Offsets Trading has grown exponentially in recent years, to > USD 12 billion value of Agreements in 2025. Please see a small deck of slides I put together.

Nations like Russia, Brazil, Indonesia, have huge potential for CDR (CO₂ Direct Removal) as well as CCUS projects. India & China have demand for Carbon offsets. Hence, it makes sense to build an intra BRICS Voluntary Carbon Market, with emphasis on Carbon Offsets

5) This approach will be in line with the proposed EGEWG sub-theme “**Circular Carbon Economy**”: which encompasses Waste to Energy, CO₂ Direct Removal (NBS, Biochar, etc) and CCUS (engineered mineralization as well as geological storage of CO₂), with activities defined as

Develop harmonised MRV, accounting, and standards for CCUS, waste-to-energy, biochar, and nature-based solutions across BRICS.

Launch BRICS pilot corridors for CCUS and carbon utilization in hard-to-abate sectors such as cement, steel, refining, and chemicals.

Enable cross-border learning and finance for circular carbon technologies through pooled green funds, concessional finance, and risk-sharing instruments.

Integrate circular carbon solutions into urban planning and industrial policy, ensuring alignment with waste management, clean energy, and climate goals

BRICS Hub for Voluntary Carbon Markets (VCM)

Updated Version for the 2026 BRICS Working Group

1. Executive Overview

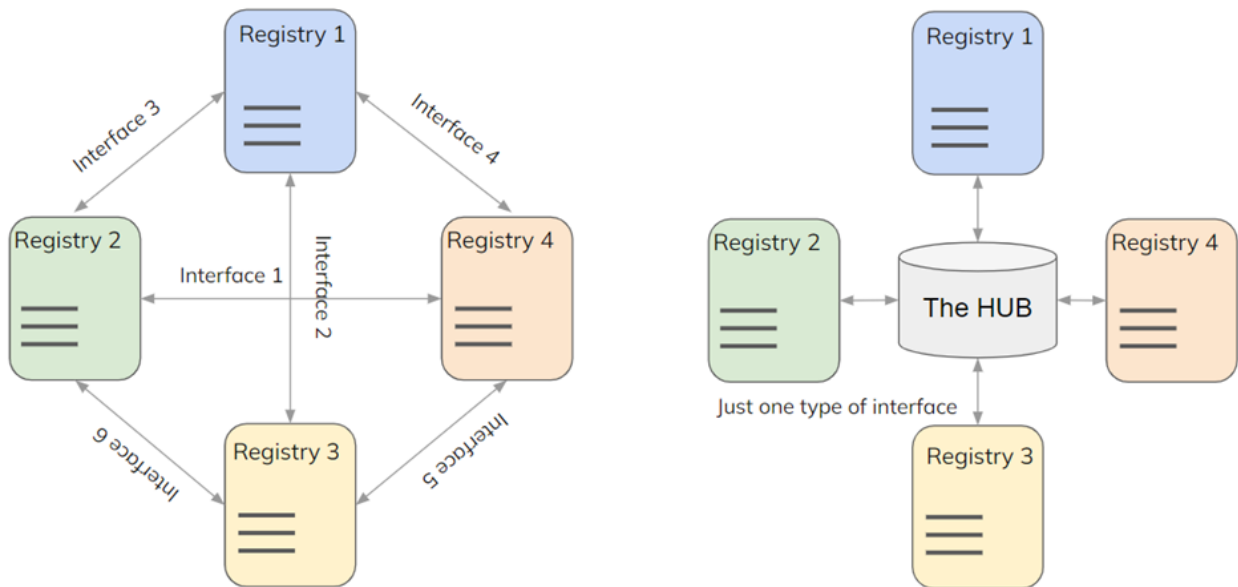
Since its launch under the Russian Presidency in 2024 and subsequent endorsement under the Brazilian Presidency in 2025, the BRICS Hub initiative has evolved into a proposal for infrastructure aimed at ensuring communication and data exchange between BRICS national VCM operators. In 2026, alongside the Indian Presidency's proposal for the BRICS VCM Network, the Hub initiative is positioned as the technical infrastructure that enables the practical execution of cross-border carbon credit transfers within the BRICS framework.

2. Core Function of the BRICS Hub

The Hub is an infrastructure mechanism that allows national carbon registries to communicate without compromising their independence.

- **Unified Interface:** Companies can conduct cross-border transactions through their own national registry without needing to open accounts in every foreign registry.

- **National Sovereignty:** Each BRICS nation maintains full control over its own regulatory protocols, standards, and legislative environment.
- **Interoperability:** The Hub ensures that credits generated in one country are acknowledged and validated in another through standardized verification protocols.



Addressing the Standards Challenge: The development of common standards for carbon projects and offset validation should be handled simultaneously with infrastructure development. The Hub provides a practical focal point for these discussions, enabling the Common Expert Body (CEB) to establish standardised protocols and best practices that will bring the diverse national markets into line.

3. Key Benefits

The implementation of the BRICS Hub provides several strategic advantages for member nations and market participants:

Benefit Category	Description
Transaction Costs	Significantly reduces administrative and operational costs for companies engaged in cross-border trading.
Transparency	Ensures a secure and auditable record of all credit transfers between national systems.

Benefit Category	Description
Integrity	Strictly prevents double counting of carbon units through synchronized cancellation and issuance protocols.
Flexibility	Supports the gradual onboarding of national registries as they reach technical maturity.
Market Liquidity	Facilitates a larger pool of participants, leading to more robust price discovery and market stability.

4. Alignment with International Standards (Article 6)

The BRICS Hub supports the aims of Article 6 of the Paris Agreement by encouraging collaborative approaches through Internationally Transferred Mitigation Outcomes (ITMOs). It provides nations with a reliable technical foundation to help them meet their Nationally Determined Contributions (NDCs) while supporting sustainable development. Ultimately, this infrastructure will facilitate interaction with broader international mechanisms, including UN registries.

5. Implementation Strategy for 2026

The implementation of the Hub follows a phased approach, prioritising foundational dialogue before full-scale deployment.

2026: Technical and institutional dialogue. The primary objective is to initiate discussions on specific topics between national voluntary carbon registry operators and the relevant competent authorities. These discussions will focus on technological choices, governance arrangements and practical implementation models.

Future Phases (2027): Pilot. In subsequent years, the focus will shift to developing a meta-registry or demo version of the Hub. This platform will provide a consolidated view of the information contained within national registries and publish data on the carbon offsets and VCM deals available between companies across the BRICS nations, without requiring immediate unit transfers.

2028: Full platform launch. The Hub will be deployed as a comprehensive platform to facilitate carbon offset trading, initially supporting a common market for early-adopter nations.

Next Step: The Russian side invites fellow Working Group members to facilitate introductions to their respective national carbon registry operators or relevant regulatory bodies to begin these practical discussions.

6. Synergy: The Network and The Hub

The **BRICS VCM Network** and the **BRICS Hub** are complementary, non-contradictory initiatives designed to build a robust carbon market ecosystem:

- **BRICS VCM Network:** Serves as the **dialogue platform** for businesses, industry experts, and regulators to coordinate standards and share best practices.
- **BRICS Hub:** Acts as the **technical backbone**, providing the digital infrastructure for interaction between national registries.

By integrating these initiatives, BRICS countries can ensure that policy discussions (Network) lead directly to seamless technical execution (Hub).

Energy Attributes Certificates (EACs) Systems in BRICS Countries

1. Context and relevance

Energy Attributes Certificates (EACs) systems are the key instrument for verifiable accounting of renewable electricity generation and consumption. They link physical generation with end-user claims under Scope 2 of the GHG Protocol, corporate sustainability disclosure, product labelling, and supply-chain claims. National and voluntary certification systems are already operating in most BRICS+ jurisdictions: Brazil, Egypt, India, China, the UAE, Russia, and South Africa.

2. Issues requiring joint analysis

Although national systems are in place, they were developed independently and differ in legal basis, institutional governance, and rules for issuance, transfer, and cancellation of certificates. At present, no consolidated comparative overview of BRICS systems exists: information on national regimes is fragmented, methodological differences are not systematised, and compatibility with international reporting practice has not been assessed at the bloc level. Addressing this gap requires coordinated analytical work within the Working Group.

3. Objective of the joint work

To prepare a technical report presenting a comparative analysis of certification systems across BRICS countries, identify common functional principles and key differences in design and governance, and assess the alignment of national systems with international reporting and disclosure practice.

4. Expected outputs

The joint work will deliver: country profiles of certification systems; a comparative matrix covering key parameters; and a gap analysis. All materials will be consolidated into a single technical package for expert review.

5. Assessment criteria

The analysis applies the following criteria:

- Legal basis and institutional governance
- Verified metering and data quality
- Unique issuance and registry integrity
- Rules for transfer, retirement, and cancellation
- Mechanisms for double-counting prevention and disclosure
- Compatibility with the GHG Protocol and comparable international standards

6. Roadmap

Stage	Timeline	Description
Stage 1	Q4 2026	Technical report development: collection of national system descriptions from participating countries and drafting of findings.
Stage 2	Q1–Q3 2027	Working Group review and revision: expert review, comments from participating countries, and refinement of analytical conclusions.
Stage 3	2028	Report approval and publication: final endorsement by the Working Group and release of the technical report.

7. Rationale for initiating the work

The timely start of joint work will ensure the development of a transparent comparative picture of EAC systems in BRICS countries, which will be in demand among regulators, businesses and the expert community. The outcome of this work will facilitate full and fair accounting of the energy and carbon components in bilateral trade and in the investment sphere. The capabilities of the certification system will make it possible to combine, in a more flexible way, the growing global requirements for carbon regulation with the need to develop the power system on the basis of various technologies. Such cooperation between BRICS countries will become an important condition for adapting to the new requirements of international trade, as well as for simplifying and more effectively developing economic relations between our countries.

8. Proposal to the Working Group

It is proposed to approve the initiation of joint work on the preparation of a technical report on Energy Attributes Certificates systems in BRICS countries in line with the presented roadmap.

The initiative is coordinated by the Russian Chapter. draft, April 2026

FROM: BRICS ENERGY, GREEN ECONOMY AND CLIMATE WORKING GROUP

CONCEPT NOTE: BRICS VOLUNTARY CARBON MARKET STRATEGY

1. CARBON MARKET ECOSYSTEM

Carbon markets have evolved over the past two decades from project-based mechanisms under the Kyoto Protocol to sophisticated domestic, voluntary and international trading systems that mobilise private capital for climate action. **Modern carbon markets provide financial incentives for reducing, avoiding or removing greenhouse gas emissions by creating tradable environmental assets** that can be purchased by governments and businesses to meet regulatory obligations or voluntary net-zero commitments. Increasingly, high-integrity carbon markets are underpinned by internationally recognised standards, robust measurement, reporting and verification (MRV) systems, transparent registries, digital trading infrastructure and strong governance frameworks. As countries implement the Paris Agreement and Article 6, carbon markets are expected to play an increasingly important role in financing the global low-carbon transition and supporting sustainable development.

SA has progressively developed the Global South's most sophisticated carbon market ecosystems. It has **transitioned towards a hybrid model where VCMs complement domestic climate policy** and is now anticipating Article 6. Further, its Carbon Offset Mechanism allows regulated emitters to purchase approved offsets from qualifying projects; which creates a **domestic-buyer base** independent of international voluntary demand. The South African Reserve Bank (SARB) is part of the Sustainable Banking and Finance Network and is a member and supervision of the Network of Central Banks' – Greening the Financial System. An overview of this journey follows:



2. VOLUNTARY CARBON MARKETS (VCM)

VCMs are platforms where market participants e.g. individuals, companies and nations **voluntarily trade carbon credits to offset their carbon footprint**. Unlike compliance markets, participation is **driven by choice and responsibility**. Wiling buyers purchase credits from projects that reduce or remove greenhouse gas emissions, demonstrating commitment to sustainability. Thus, VCM's effectively monetising environmental benefits through carbon credits by allowing sellers to improve revenues or offset operational costs; ultimately improving

their financial attractiveness. Moreover, the credits contribute towards national **carbon emissions reduction goals and Nationally Determined Contributions (NDCs)**.

2.1. Paris Agreement – Article 6

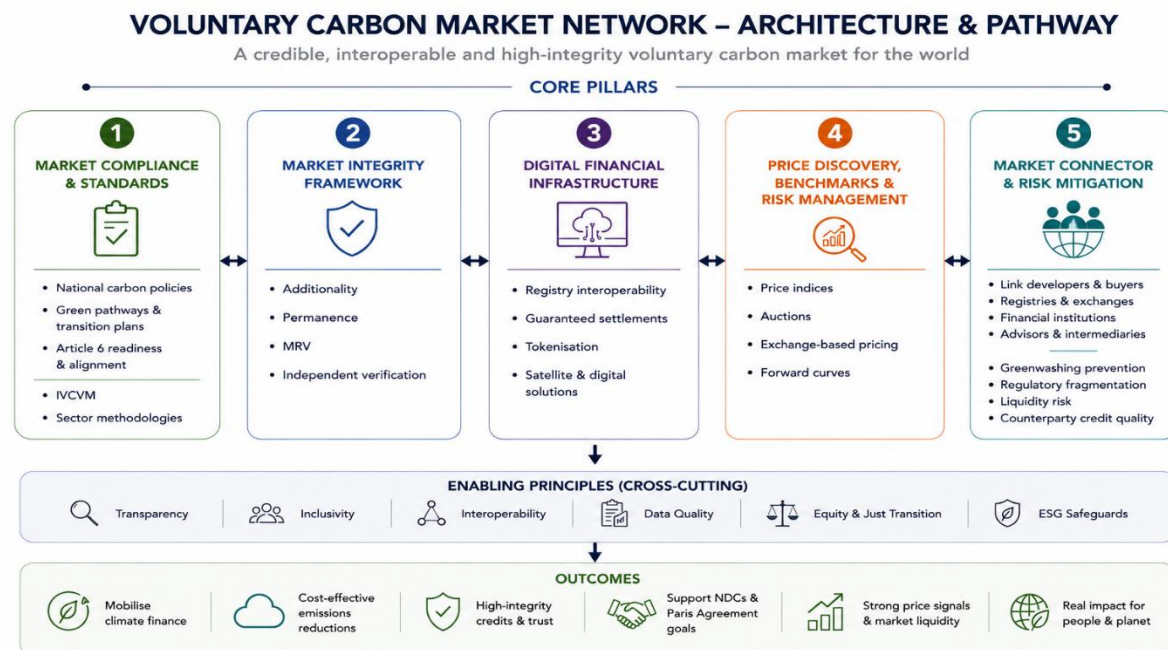
Article 6 regulates international carbon markets and cooperation between countries to meet their climate targets (NDCs) and provides “guiding principles” on carbon trading. It allows countries to: 1) Bilaterally and multi-laterally trade emissions reductions 2) Cooperate on climate projects and 3) Lower the cost of achieving NDCs. **The BRICS VCM should align with Article 6**, enabling the generation and transfer of internationally transferable mitigation outcomes (ITMOs), subject to host country authorisation and corresponding adjustments where applicable. By incorporating Article - 6 principles, the envisioned BRICS VCM Network can evolve into a **credible globally integrated system to scale finance and guardrail against market failure**.

2.2. Carbon Border Adjustment Mechanism (CBAM)

Effective 2026, the EU will impose tariffs on carbon-intensive imports (steel, cement and aluminium), compelling foreign producers to align with EU climate goals to access the EU’s €16 billion import market. In 2025, the BRICS nations jointly rejected CBAM, due its perceived discriminatory and detrimental impact on their economic development. The policy violates Paris Agreement, WTO and UNFCCC principles; the later which calls for “common but differentiated responsibilities”; which creates tensions between climate ambition and equity.

2.3. VCM Risks and Architecture Foundation

Key risk attributed to VCM include the following: Market Integrity, Greenwashing, Regulatory and compliance, Liquidity, Counterparty credit quality, Cyber-attacks and fraud, ESG Safeguards. The establishment of a VCM requires more than a trading platform, it requires a robust market architecture that promotes integrity, transparency, interoperability and investor confidence. The proposed framework is built around five interconnected pillars covering market compliance and standards, market integrity, digital financial infrastructure, price discovery, and market connectivity and risk management. Together, these pillars provide the institutional, regulatory and financial foundations needed to mobilise climate finance, facilitate cross-border carbon trading, support Article 6 cooperation, and position BRICS as a leading participant in high-integrity global carbon markets as follows:



3. BRICS' VOLUNTARY CARBON TRADING PLATFORMS

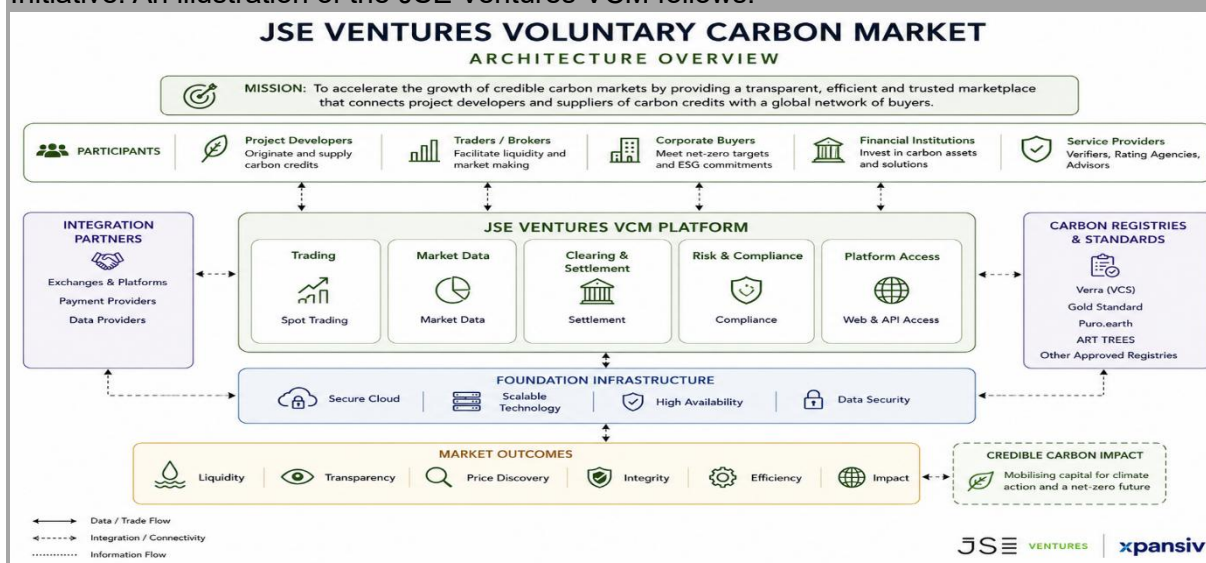
The BRICS market mainly has **fragmented domestically carbon trading systems** at different sophistication levels; that will **benefit from alignment and integration**. [Brazil primarily relies on implicit carbon pricing but has enacted law to establish a Brazilian Emissions Trading System (SBCE). Russia's strategy is voluntary corporate emission reporting and carbon unit issuance schemes. India has an evolving Carbon Credit Trading Scheme (CCTS); a domestic industrial carbon market.] China relies on its National Emissions Trading System (ETS) is a cap-and-trade system that regulates green-house gas emissions; by imposing caps on emitters above 25 000 tons of CO₂ p.a. and requires reporting from emitter above 10 000 CO₂ tons p.a. The ETS was established in 2021 and regulates more than 3 300 industrial companies and provides tradeable free allowances (based on out-put-based pricing system) to power and hard to abate sectors. Use of proceeds are not prescribed and local carbon setting is also undertaken. In 2024, China launched its Certified Emissions Reduction Scheme (CCER programme) which operates a registry to trade price-based instruments. Its MVR framework includes oversight on Provincial level and environmental authorities e.g. Ministry of Ecology and Environment.

Case Study: Johannesburg Stock Exchange (JSE) Ventures VCM

The VCM is one of SA's most significant developments; it aims to **assist the African to accelerate carbon offset projects towards Net Zero commitments**. The market enables private and public participants to buy carbon credits to offset carbon emissions and conversely provides a mechanism for project developers to sell carbon credits to gain returns from carbon reduction and avoidance projects. The credits are sold to industries which are otherwise unable to reduce carbon emissions to meet their carbon tax obligations. Thus, it **connects African buyers with internationally verified credits and allows for global offsetting**.

The JSE has partnered with Xpansiv, an infrastructure provider for global environmental markets, to provide a centralised online marketplace with an efficient and transparent ecosystem to exchange carbon offset credits and international renewable energy certificates (I-REC); registered under global carbon standards. The market was established in 2023 following key drivers that include South African Carbon Tax Act, 2050 Net Zero Targets,

Nationally Determined Contributions (NDCs) and alignment with the African Carbon Market Initiative. An illustration of the JSE Ventures VCM follows:



It is recommended that the **Global Working Group makes a formal collaboration request to the JSE VCM**. A collaboration between the JSE Ventures VCM and BRICS will yield the following benefits: institutional **governance and digital backbone**; global **market access and liquidity** through Xpansiv; centralised trading platform delivering **aggregation, transparency and credibility**; **Regional integration** across global registries; **clearing and settlement functions**; **compliance convergence** by integrating voluntary and compliance markets; **funds mobilisation**; and **global South-led carbon pricing** solutions.

3.1. VCM Trading Instruments / Products

VCMs facilitate the trading of environmental commodities that enable organisations to meet climate commitments, demonstrate renewable energy consumption and support the financing of emissions reduction and removal projects. The most widely traded products include carbon offset credits and renewable energy certificates, each serving distinct but complementary roles in the transition to a low-carbon economy.

3.1.1. Offset Carbon Credits

Carbon offset credits represent a verified reduction, avoidance or removal of greenhouse gas (GHG) emissions. One carbon credit is generally equivalent to one tonne of **carbon dioxide equivalent (1t CO₂e)** that has been independently measured, verified and registered under an internationally recognised carbon standard. Project developers generate credits that are purchased voluntarily by corporates or used for regulatory compliance, where permitted.

Case Study 2: City of Cape Town – Carbon Credit Trading (Landfill Gas Extraction and Utilisation Project)

During 2024, the City earned ca R35 million from its first carbon credit auction through the JSE. The carbon credits were generated by reducing methane gas emissions at City landfill sites (Bellville, Coastal Park and Vissershok) via waste-to-energy initiatives. The gas-infrastructure has also led to a reduction in bulk electricity from Eskom. Proceeds from the sale were ring-fenced and reinvested to develop additional W-t-E plants or fund pollution reduction and landfill diversion on community level. The landfill-based carbon credits were registered under the CDM and complied with ISO standards as well as technical and legal specifications of a Programme of Activities (PoA). The carbon emission reductions were

independently verified by a UN. The credits contributed towards the City's carbon emissions reduction goals and SA's NDCs. Regional demand for carbon offset exceeded demands, with high emitting corporations bidding competitively to offset their liabilities under the SA Carbon Tax Act. The auction was 3,4 times oversubscribed.

3.1.2. International Renewable Energy Certificates (I-RECS)

International Renewable Energy Certificates (I-RECs) provide proof that **one megawatt-hour (1 MWh)** of electricity has been generated from an eligible renewable energy source and injected into the electricity grid. Unlike carbon credits, I-RECs do not represent avoided emissions but certify renewable electricity generation and consumption. Corporate electricity consumers purchase and retire these certificates to demonstrate renewable electricity consumption for sustainability reporting and Scope 2 greenhouse gas accounting.

Case Study: Empower Trading – Solar Capital I-REC transaction

To lower its market-based Scope 2 emissions and mitigate domestic Carbon Tax, an industrial manufacturer in Flexiton – KwaZuluNatal Province, entered into a consumer power purchase agreement (cPPA) with Empower Trading, a 3rd party electricity trader which entered into a back-to-back generator PPA (gPPA) with Solar Capital. Solar is an independent power producer (IPP) which operates a 10MW solar PV farm in De Aar – Northern Cape Province. The power was wheeled through Eskom's Wheeling Framework. The project registered 20 000 MWh via the I-REC Standard Foundation and awarded 20 000 I-RECs, valued at ca R0.025 per kWh. The I-RECs were transfer to the manufacturer's registry account, subsequently redeemed and audited; using Carbon Disclosure Project (CDP).

4. CARBON TAX AND GREEN FUNDING MECHANISMS

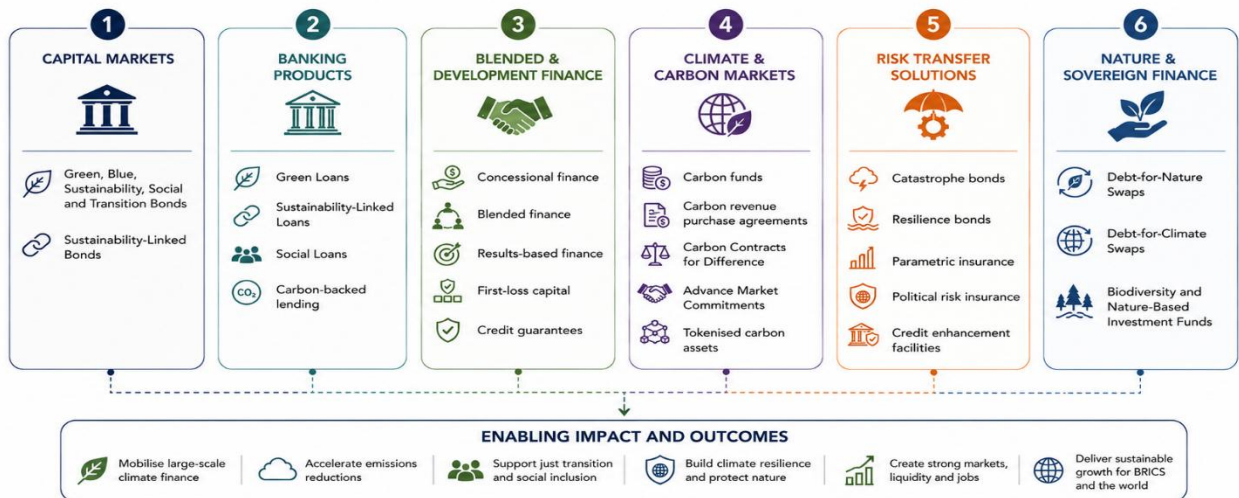
Few BRICS nations currently have formal, unifying carbon tax legislation. India manages carbon pricing through excise duties on fossil fuels and coal cess (general sale tax). Brazil, Russia and China have no national carbon tax frameworks. Finance remains a major barrier to action but recent efforts to reform global financial architecture offer promising pathways for more equitable and affordable climate finance. SA was the first African country to introduce Carbon Tax in 2019, which currently stands at ca. ZAR190 (ca. US\$12) per CO₂ ton.

5. GREEN FINANCING INSTRUMENTS

Climate change is destabilising developing economies with **the cost of decarbonation exceeding national budgets**, an estimated **US\$2 trillion** is required for EMDC to meet climate and SDG goals. Climate financing for developing countries is a debt trap as financing comes in the form of limited grant capital and unaffordable debt financing. In 2023, 75% for SA's climate finance was debt-based and 85% of its JET-IP commitments are debt. Thus, debt reform and non-traditional climate finance is required. There is a call for **global financial reform and review of national climate financing mechanisms including green taxonomies, carbon pricing, central bank regulation and affordable climate finance solutions**. Emerging climate finance solutions include the following funding instruments: 1) capital markets instruments 2) investment banking products 3) blended and development finance 4) climate and carbon funding 5) risk transfer for environmental disaster and nature and 6) nature and sovereign finance for climate adaptation:

BRICS CLEAN ENERGY TRANSITION FINANCING ECOSYSTEM

A Comprehensive Toolkit for Mobilising Climate and Sustainable Investment



Case Study: SA Municipal Green Bonds

In 2017, the City issued the nation's first green bond to finance water and sanitation projects; which received a Moody's "excellent GB1" rating. EMDC's represent <20% of green bond issuances. City of Johannesburg followed in 2019. Many of SA's metropolitan municipalities have adopted Green Bond Frameworks. In 2026, National Treasury issued a regulation governing use of proceeds for sustainable projects (green, sustainable and social) aligned with ICMA.

Case Study: Cape Water Nature - Based Bond

The project is a global landmark R2,5 billion (US\$150 million) blended financing linked to investor returns to verified ecological and environmental restoration. It signalled to the market that natural capital has entered the mainstream finance in SA. The use of proceeds was to clear invasive alien plants to restore waterflow from catchments in the Western Cape; to increase stream flow into storage dams that support critical agri-food value chains. The Nature Conservancy was the implementing partner, Anchor investors include IFC, FDS Africa (UK) and Aluwani (SA infrastructure fund). Outcome-based funders included FirstRand Foundation and Development Bank of Southern Africa. Rand Merchant Bank / FNB was the debt issuer and project agent, with independent verification by Conservation Alpha. The bond is JSE listed.

Case Study: Debt-for-Nature Swaps – Seychelles

In 2016, the Nature Conservancy (TNC) provided a US\$22 million financing (grant and loan) to Seychelles Conversation and Comate Adaptation Trust (SeyCCAT). SeyCCAT in turn issued a low-interest loan to the Seychelles Government (Debtor) as Debtor at a 6.5%V discount, which was used to repay existing more expensive debt from the Paris Club (Creditor) comprising of France, UK, Belgium and Italy. Seychelles moved from zero (<1%) to 30% marine protection metrics, covering 410 000 km of ocean. It resulted in coral reef protection, species protection and fisheries recover. Seychelles issued its maiden SOvereign Blue Bond in 2018. No certification was issued but the project met the nations own marine spatial plan, protection laws and zoning policies. Funding allocations can be varied by TNC as part of loan covenants tied to environmental compliance. The project achieved participation of local conservation and communities.

6. BRICS VCM NETWORK CONSIDERATIONS

It is recommended that the BRICS EGE Working Group establish a BRICS Voluntary Carbon Market (VCM) Network to **connect existing national carbon markets, registries, exchanges** and market participants through a common, high-integrity framework aligned with international best practice such as Paris Agreement - Article 6. The Network should **leverage existing market infrastructure** to promote interoperability, market integrity, price discovery, climate finance mobilisation and private sector participation. This collaborative approach will **accelerate cross-border carbon trading, support high-quality mitigation projects and position BRICS as a leader** in global voluntary carbon markets.

India's carbon registry, part of the Indian Carbon Market (ICM) under the Carbon Credit Trading Scheme (CCTS), is managed by the **Grid Controller of India (GCI)** as the registry operator, while the **Bureau of Energy Efficiency (BEE)** serves as the Administrator.

GCI maintains the secure national registry, tracking entities and carbon credits, with oversight from the National Steering Committee (NSC-ICM) co-chaired by Ministry of Power and Environment Secretaries.

Key Bodies & Roles:

- **Grid Controller of India (GCI)**: The technical operator, managing the registry's database, credit accounts, and facilitating transactions.
- **Bureau of Energy Efficiency (BEE)**: The overall scheme administrator, creating rules, standards, and overseeing the market's functioning.
- **National Steering Committee (NSC-ICM)**: The highest authority, guiding policy, targets, and guidelines for the carbon market.
- **Central Electricity Regulatory Commission (CERC)**: Regulates trading activities on exchanges, preventing fraud.

In essence, GCI runs the registry platform, BEE manages the overall system, and the NSC-ICM provides top-level governance for India's carbon market.

Carbon Credit Trading Scheme (CCTS) - Indian Carbon Market registry is managed and operated by the Grid Controller of India (GCI), with BEE being scheme administrator, pertains to the national

Voluntary Carbon Market in India, registries are managed by private and non-governmental entities, that are aligned with international

standards (Verra, Gold Standard, etc), eg **Carbon Registry – India**, established by Network for Certification and Conservation of Forests (NCCF),

India's official carbon registry for the national Carbon Credit Trading Scheme (CCTS) is managed and operated by the Grid Controller of India (GCI). The registry is also referred to as the Indian Carbon Market (ICM) registry.

ROLES AND RESPONSIBILITIES: The institutional framework for the Indian Carbon Market (ICM) involves several key government bodies:

REGISTRY OPERATOR: The Grid Controller of India (GCI) is responsible for the operational management of the electronic registry platform. Its functions include:

- Registering entities in the carbon market.

- Maintaining accounts of all issued carbon credits.

- Facilitating the transactions of credits on the trading platform.

- Functioning as the meta-registry for the country and maintaining a secure database.

SCHEME ADMINISTRATOR: The Bureau of Energy Efficiency (BEE), under the Ministry of Power, acts as the administrator of the CCTS. It is responsible for developing technical procedures, identifying sectors, issuing the Carbon Credit Certificates (CCCs), and overseeing monitoring, reporting, and verification (MRV) activities.

OVERSIGHT AND POLICY:

A National Steering Committee for the Indian Carbon Market (NSCICM) provides overall guidance and oversight.

The Central Electricity Regulatory Commission (CERC) regulates the trading activities of the carbon credits on the power exchanges.

The Ministry of Environment, Forest and Climate Change (MoEFCC) formally notifies the GHG emission targets and designates the obligated entities under the Environment Protection Act, 1986.

VOLUNTARY REGISTRIES

For the voluntary carbon market in India, several private and non-governmental entities also operate registries that align with international standards (like Verra or Gold Standard).

CARBON REGISTRY – INDIA (CR-I), an initiative by the Network for Certification and Conservation of Forests (NCCF), which facilitates voluntary project listing, tracking, and trading.

Carbon Registry – India, <https://cri.nccf.in>

Network for Certification and Conservation of Forests (NCCF), came into existence in January 2015 as a registered society, comprising experts, professionals, and stakeholders of natural resources, engaged in the development of India specific and globally benchmarked sustainability standards, as its core working area, apart from working for policy advocacy, training and capacity building through multi-stakeholder engagement.

NCCF is a Member of the PEFC Alliance from India, a member of IUCN, and has got Observer Status with the UNCCD as a Civil Society Organization.